A computational model of the evolution of honeybee waggle-dance communication

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1 Introduction

Honeybee waggle dances provide spatial information about resources away from the nest. One ecological hypothesis is that such communication is favored when food is difficult to discover independently but remains useful after a worker has found it. Empirical and theoretical studies suggest that the value of dance information and recruitment depends on the spatiotemporal distribution of food resources, including patch density, patch size, reward variability, and habitat [7, 3, 4, 1, 2, 6, 5]. We begin with a deliberately simple model of directional communication on an exposed horizontal comb, where dance angle can map directly onto food direction.

2 Method

Colonies are the reproducing entities, while workers are behavioral agents. Each colony has seven heritable mean traits: directional precision investment, called directional bias in the simulation code and tables, which controls how strongly a dancing worker's angle is centered on the intended direction; receiver attention, which controls how likely workers are to follow an available dance; sender transposition, which controls how much a dancer maps food direction onto a gravity-referenced dance angle; receiver transposition, which controls how much dance followers interpret the dance as gravity-referenced rather than direct pointing; comb tilt and comb orientation, which define the dance plane in the world; and search limit, which controls how far a worker searches along its chosen direction before giving up. Individual workers are sampled around the colony means using stable within-colony variation. Each dance also includes transient production and interpretation noise.

The directional-precision trait should be read as a costly investment in producing directionally informative dances, not as a pure measure of neural precision, motivation, or dance duration. In this model, b sets the angular concentration of a produced dance and contributes to the per-dance cost paid by the colony. Dance cost is represented as a baseline cost plus a precision-dependent cost. The baseline component captures time, metabolic effort, or opportunity cost of dancing regardless of directional precision, whereas the precision-dependent component captures the additional cost of making the dance directionally informative. The reported simulations set the baseline dance cost to 0 and therefore include only the precision-dependent component. Thus the results show the evolution of costly signal precision under that simplified cost schedule, not the separate evolution of intrinsic accuracy and signaling effort.

More concretely, a colony is parameterized by mean directional precision investment $b \in [0, 1]$, mean receiver attention $a \in [0, 1]$, mean sender transposition $s \in [0, 1]$, mean receiver transposition

$r \in [0, 1]$, and mean search limit $\ell \geq 0$. It also has a comb tilt $t \in [0, 1]$ and a comb orientation ϕ. Orientation can be treated either as a circular direction or, in constrained probes, as an axial variable where ϕ and $\phi+180^\circ$ represent the same comb plane. Each worker receives stable individual deviations from these means, sampled from Gaussian noise with standard deviation 0.08 for unit-interval traits. For search limit, the same scale is multiplied by the maximum search distance. All traits are clipped to their allowed ranges. These deviations represent within-colony worker variation, but are not themselves heritable in the current model. At reproduction, only the colony-level mean traits are inherited, with Gaussian mutation of standard deviation 0.07, again scaled by maximum search distance for the search-limit trait. Sender and receiver transposition mutations can be correlated by a configurable coupling parameter; comb-orientation mutations use a separate circular mutation scale. Comb tilt can also use its own mutation scale when an experiment supplies a geometric precondition rather than asking tilt to evolve.

Each foraging episode generates a set of food sites in the world. A food site has a direction, distance from the nest, angular width, value, and capacity. Workers act sequentially within an episode. If at least one dance is available, a worker follows a randomly chosen dance with probability given by its receiver attention. Otherwise it searches independently in a uniformly random direction. In both cases, the worker searches along the chosen direction up to its individual search limit. A food site is found only if it lies within the searched angular window and its distance is no greater than the worker's search limit. Independent searchers that find food add a noisy dance for the discovered site, so scouts are not assigned in advance: a scout is simply a worker that found food independently. This makes both background discovery and recruitment emergent properties of the food distribution and the evolving worker traits. This design follows the logic of previous ecological models in which recruitment benefits depend on the availability and discoverability of resource patches rather than on communication having a fixed payoff advantage [4, 1].

Comb tilt and orientation define a three-dimensional comb plane. Direct pointing projects the horizontal food direction onto that plane, so its available cue strength depends on both the tilt and the orientation of the comb relative to the food direction. Gravity-referenced mapping uses the projection of gravity onto the comb plane together with an episode-level sun azimuth. Gravity provides no in-plane reference on a horizontal comb and becomes a stronger cue as the comb becomes vertical. The sender transposition trait blends between direct pointing and sun/gravity-referenced encoding, and the receiver transposition trait blends between the corresponding decoding rules. Each episode samples the sun azimuth from a configurable daytime arc.

The dancer's angle is sampled from a circular von Mises distribution centered on the encoded direction. The concentration parameter is $14b$, so workers with low directional precision investment produce nearly random angles, while workers with high directional precision investment produce angles concentrated around the intended direction. Additional dance-production noise with standard deviation 0.18 is added to the angle. A worker follows an available dance with probability a; if it follows, it decodes the signaled direction with interpretation noise of standard deviation 0.12. If it does not follow a dance, it searches in a uniformly random direction. A worker succeeds when its search direction falls within the angular width of an available food site whose distance is no greater than the worker's search limit.

After many episodes, each colony receives a payoff from collected food minus travel, communication, and attention costs. Daughter colonies are sampled in proportion to payoff, and colony-level traits mutate by Gaussian noise. We track the mean directional precision investment, receiver attention, sender and receiver transposition, comb tilt, comb-orientation alignment, and search limit, plus forager success rate across generations.

In each episode, collected food is the sum of the values of food sites found by workers. The current simulations use food value 1.0. Workers also pay a travel cost proportional to distance:

successful workers pay the cost of reaching the found food site, while unsuccessful workers pay the cost of searching up to their limit. The default travel cost is 0.02 per distance unit, so even the longest default trip remains profitable if it finds food, but long unsuccessful searches are costly. The colony then pays a communication cost of $c_0 + 0.02b_d$ for each dance added by an independent finder, where c_0 is the baseline dance cost and b_d is that dancer's directional precision investment. In the experiments reported here, $c_0 = 0$. The colony also pays an attention cost of 0.01 for each worker that follows a dance. Thus more precise directional signaling, more dance-following, and longer search effort can increase food collection, but they are not free. The current model can also include a configurable benefit proportional to comb verticality, which represents non-communication selective pressure for vertical combs. The parameter "workers per colony" controls the pool from which active foragers are sampled, and therefore the amount of within-colony worker heterogeneity, but payoff opportunity is set more directly by the number of foraging episodes per colony and the number of foraging attempts per episode.

3 Results

3.1 Horizontal comb

We compared three simple food distributions over ten random seeds each (seeds 101–110). The "hard" condition had one narrow food site per episode, the baseline condition had two moderately wide food sites, and the "easy" condition had many broad food sites. Table 1 reports the generation at which mean directional precision investment first reached 0.30, plus final generation trait and success values.

Each run used 60 colonies, 80 workers per colony, 30 generations, 50 foraging episodes per colony per generation, 12 foraging attempts per episode, and horizontal-comb tilt $t = 0$. Food-site distances were sampled uniformly from 1.0 to 8.0, maximum search distance was 8.0, travel cost was 0.02 per distance unit, baseline dance cost was 0, the precision-dependent dance-cost coefficient was 0.02, and food-site value was 1.0. All three conditions used food-site capacity 6. The conditions varied only the number and angular width of food sites: hard used one site of width 0.08, baseline used two sites of width 0.20, and easy used eight sites of width 0.50. The hard condition therefore makes independent discovery unlikely, while the easy condition makes independent discovery likely even without using the dance.

Condition	Reached	Reach gen.	Bias	Attention	Search	Success	Payoff
Hard	6/10	23.2	0.271	0.357	3.070	0.009	0.001
Baseline	10/10	10.2	0.850	0.852	7.155	0.195	0.742
Easy	7/10	21.4	0.352	0.335	6.917	0.703	7.320

Table 1: Food-distribution experiment under the current dynamic recruitment model. "Reached" is the number of seeds in which mean directional precision investment reached 0.30 by generation 30. "Reach gen." is averaged over reached seeds only. Bias is the table label for directional precision investment; the reported bias, attention, search limit, success, and payoff values are final-generation means over all ten seeds.

The result matches the ecological prediction qualitatively, but with a sharper distinction than in the earlier fixed-scout model. In the hard environment, food is so rare and narrow that independent discoveries seldom seed dances; final success and payoff remain close to zero, and communication evolves only in some seeds. In the baseline environment, independent discovery is sufficient to seed recruitment cascades, and directional precision investment, receiver attention, and search limit all rise strongly. In the easy environment, success and payoff are high, but directional precision investment and receiver attention remain much lower than in the baseline case because food is often found without precise communication.

3.2 Vertical comb

We also compared horizontal, tilted, and vertical combs while holding the baseline food distribution fixed. Comb tilt was set to $t = 0$, $t = 0.5$, or $t = 1.0$, with ten seeds per condition. Table 2 summarizes the result after 30 generations.

Comb	Reached	Reach gen.	Bias	Attention	Sender	Receiver	Success	Payoff
Horizontal	10/10	10.2	0.850	0.852	0.278	0.397	0.195	0.742
Tilted	9/10	22.3	0.455	0.740	0.659	0.586	0.146	0.223
Vertical	5/10	21.0	0.277	0.337	0.376	0.352	0.099	0.013

Table 2: Comb-tilt comparison using the baseline food distribution and ten seeds. Sender and receiver denote final mean sender and receiver transposition traits.

As expected, direct horizontal pointing performs best when the comb is horizontal. Increasing comb tilt weakens the existing direct mapping and slows the evolution of directional signaling. In the tilted condition, sender and receiver transposition both rise, indicating selection toward the gravity-referenced mapping, but payoff remains lower than in the horizontal condition. In the fully vertical condition, communication remains weak after 30 generations: only half of the seeds reach the directional-precision threshold, and final payoff is close to the minimum floor.

3.3 Tilt-geometry calibration

After replacing the scalar comb-tilt degradation with explicit comb-plane geometry, we ran a calibration grid over initial comb tilt and the vertical-comb benefit. These runs used the baseline food distribution, the daytime sun arc from 0° to 180° centered at 90°, and ten seeds (101–110). Success is the fraction of foraging attempts that found food. Payoff is the final mean colony payoff after food rewards, search/travel costs, dance costs, attention costs, and the vertical-comb benefit. Table 3 shows that increasing the vertical benefit can preserve more vertical combs, but current settings produce a sharp tradeoff: strong verticality often comes with reduced foraging success.

The calibration suggests that the transition is path-dependent. Starting from a horizontal comb, benefits in the range 0.15–0.25 do not move colonies far from horizontal within 30 generations. Starting from a tilted comb, the same benefit range can retain intermediate tilt but with lower success. Starting from a vertical comb, benefits around 0.20–0.25 retain high verticality, but foraging nearly fails in most seeds. A separate five-seed stress test showed that a much larger benefit of 0.50 can drive high verticality from tilted starts, but at the cost of severe loss of foraging success. Thus the current model can impose a substantial architecture pressure, but communication does not yet adapt smoothly enough to make the transition robust.

3.4 Tilt orientation

We also examined the evolved orientation of the comb tilt. The current sun arc is centered at 90°, so Table 4 reports the weighted mean orientation in degrees and its offset from this sun-arc center. The within-alignment column measures how strongly colonies within a run converge on an

Initial tilt	Benefit	Final tilt	Sender	Receiver	Success	Payoff	Reached
0.0	0.15	0.142	0.178	0.183	0.195	0.735	10/10
0.0	0.20	0.162	0.192	0.208	0.195	0.762	10/10
0.0	0.25	0.151	0.229	0.196	0.194	0.751	10/10
0.5	0.15	0.373	0.225	0.230	0.137	0.535	9/10
0.5	0.20	0.445	0.282	0.271	0.137	0.558	9/10
0.5	0.25	0.330	0.208	0.206	0.154	0.601	10/10
1.0	0.15	0.737	0.341	0.339	0.090	0.313	7/10
1.0	0.20	0.919	0.291	0.282	0.001	0.093	4/10
1.0	0.25	0.917	0.288	0.333	0.002	0.127	7/10

Table 3: Ten-seed calibration of the vertical-comb benefit under the explicit tilt-geometry model. Sender and receiver are final mean transposition traits. Reached is the number of seeds in which mean directional precision investment reached 0.30 by generation 30.

orientation. The across-alignment column measures whether different seeds converge on the same absolute orientation, weighting each seed by its within-run orientation alignment.

Initial tilt	Benefit	Final tilt	Success	Within align.	Mean orient.	Sun offset	Across align.
0.0	0.15	0.142	0.195	0.518	122.8	32.8	0.473
0.0	0.20	0.162	0.195	0.489	155.1	65.1	0.492
0.0	0.25	0.151	0.194	0.478	79.0	-11.0	0.084
0.5	0.15	0.373	0.137	0.490	158.9	68.9	0.249
0.5	0.20	0.445	0.137	0.356	214.4	124.4	0.158
0.5	0.25	0.330	0.154	0.523	73.9	-16.1	0.401
1.0	0.15	0.737	0.090	0.545	251.7	161.7	0.374
1.0	0.20	0.919	0.001	0.374	263.8	173.8	0.093
1.0	0.25	0.917	0.002	0.392	266.8	176.8	0.291

Table 4: Orientation sanity check under the explicit tilt-geometry model. Mean orientation and sun offset are measured in degrees. Across alignment is the weighted across-seed orientation alignment; values near 0 indicate that different seeds choose different absolute orientations.

Orientation sometimes aligns within a run, but it does not yet show a clean, robust relationship to the daytime sun arc across seeds. Some conditions point near the sun-arc center, some near the opposite direction, and several have weak across-seed alignment. This may indicate weak or path-dependent selection on orientation. It also highlights a modeling issue: because a comb plane is unchanged by reversing its normal, comb orientation may be better treated as an axial variable, where ϕ and $\phi + 180^\circ$ are equivalent, rather than as a fully circular direction.

3.5 Constrained vertical coupling

The tilt-calibration results left open whether gravity-referenced communication is difficult because the sender–receiver code itself is hard to reach, or because colonies must evolve the vertical geometry and the sender–receiver code at the same time. To separate these possibilities, we ran a constrained probe with the geometric precondition supplied. Initial comb tilt was fixed near vertical at $t = 0.95$, comb-tilt mutation was set to 0, comb orientation was treated axially, the vertical-comb benefit was set to 0, and the baseline food distribution was otherwise retained. We then varied only the sender–receiver transposition mutation correlation and ran each condition for 120 generations over five seeds (101–105). A run was counted as reaching gravity-referenced communication when both mean sender and receiver transposition reached 0.50.

Corr.	Reached	Reach gen.	Bias	Sender	Receiver	Success	Payoff
0.0	5/5	42.6	0.878	0.864	0.881	0.192	0.680
0.3	5/5	46.2	0.847	0.913	0.898	0.192	0.694
0.6	5/5	21.0	0.882	0.890	0.897	0.199	0.754
0.9	5/5	22.4	0.863	0.907	0.903	0.194	0.723
1.0	5/5	23.4	0.836	0.901	0.901	0.191	0.677

Table 5: Constrained near-vertical coupling probe. Corr. is the correlation between sender and receiver transposition mutation increments. "Reached" counts seeds in which both mean sender and receiver transposition reached 0.50 by generation 120. Reach generation is averaged over reached seeds only.

The result is strongly diagnostic. With near-vertical geometry supplied, gravity-referenced communication is reachable in every condition. Independent or weakly coupled sender–receiver mutations still reach the gravity channel, but only after roughly 40–50 generations on average. Moderate to strong coupling accelerates the transition: correlations of 0.6–1.0 reach the threshold by about generation 21–23, while also keeping final sender and receiver transposition closely matched. The best payoff in this probe occurs at correlation 0.6. These results suggest that the earlier vertical-comb failure is not because gravity-referenced communication is intrinsically unreachable. Rather, the difficult part is the joint transition in which comb architecture, orientation representation, and the sender–receiver code must all become useful together.

4 Conclusion

The current model now separates world directions, comb-plane dance signals, comb tilt and orientation, and the sun/gravity reference. Direct pointing is available on horizontal combs without additional mapping machinery, while the sun/gravity channel becomes useful only as gravity projects into the comb plane. This supports the intended conceptual setup: a non-communication benefit for vertical combs can create pressure toward verticality, after which sender–receiver transposition may become advantageous.

The calibration runs show that this transition is not yet smooth. When verticality is weakly rewarded, colonies starting from horizontal combs remain mostly horizontal. When verticality is strongly rewarded, vertical combs can be retained, but foraging success often collapses before communication adapts. Orientation is not yet robustly selected relative to the sun arc under the circular representation. The constrained vertical-coupling probe clarifies this picture: once near-vertical geometry is supplied and orientation is treated axially, gravity-referenced sender–receiver transposition evolves reliably, especially when sender and receiver transposition mutations are moderately or strongly coupled. The next modeling steps are therefore to use the axial orientation representation in transition experiments from horizontal or tilted starts, and to calibrate vertical-comb benefits and mutation scales so that architecture and communication can coevolve without a collapse in foraging success.

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